

DETAILED DESCRIPTION OF THE INVENTION — DEVICE ARCHITECTURE

The following detailed description provides illustrative embodiments of a longitudinal pelvic physiology monitoring system comprising a body-positioned intraluminal physiologic monitoring device. The embodiments described are exemplary and non-limiting, and variations, substitutions, and alternative configurations are considered within the scope of the disclosure.

In various embodiments, the device comprises an insertable body configured for placement within a body lumen for physiologic measurement. The insertable body may be formed from medical-grade silicone, elastomeric polymers, thermoplastic elastomers, composite materials, or multi-durometer constructions designed to balance flexibility, structural support, and user comfort.

The device may include a proximal region, a sensing region, and a distal region. The proximal region may include a retrieval structure, handle, flange, wing, or retention geometry configured to assist insertion, orientation, and removal. Retention structures may include compliant bulbs, flexible fins, expandable structures, textured regions, or geometries configured to maintain positional stability during measurement.

The sensing region may extend longitudinally along the insertable body and may contain distributed sensing elements configured to capture physiologic interaction between tissue and device surfaces. Sensors may be arranged in rings, segments, arrays, longitudinal strips, or spatially distributed zones to enable localized measurement of physiologic conditions along different regions of the lumen.

In certain embodiments, the device may include sealed internal compartments housing electronic components including a microcontroller, signal conditioning circuitry, analog-to-digital conversion components, memory storage, wireless communication interfaces, and power subsystems. Electronics may be encapsulated using hermetic sealing, overmolding, potting compounds, or conformal coatings suitable for operation within biologic environments.

Power may be supplied using rechargeable batteries, disposable cells, inductive charging systems, magnetic charging interfaces, capacitive energy transfer, wired interfaces, or hybrid power architectures. Power management circuitry may regulate sampling frequency, sensor activation, wireless transmission intervals, and sleep modes to optimize energy efficiency during longitudinal monitoring.

The device architecture may support modular configurations in which sensing modules, electronics assemblies, or functional components may be interchangeable, upgradable, or configurable for different use cases, cost targets, or performance levels. Alternative embodiments may omit certain sensors, substitute equivalent sensing technologies, or integrate additional sensing capabilities without departing from the principles disclosed herein.

The disclosed architecture enables both session-based physiologic measurement and repeated longitudinal monitoring across multiple insertion events. Data consistency may be improved through geometry-guided positioning, calibration routines, and algorithmic compensation for placement variability.

All structural configurations described herein are illustrative examples. The invention encompasses alternative device geometries, material selections, component arrangements, and manufacturing approaches capable of achieving substantially similar functional outcomes.